# Using the One ROM for Beginners

**Manual created by Rudy's Retro Intel - 2026**

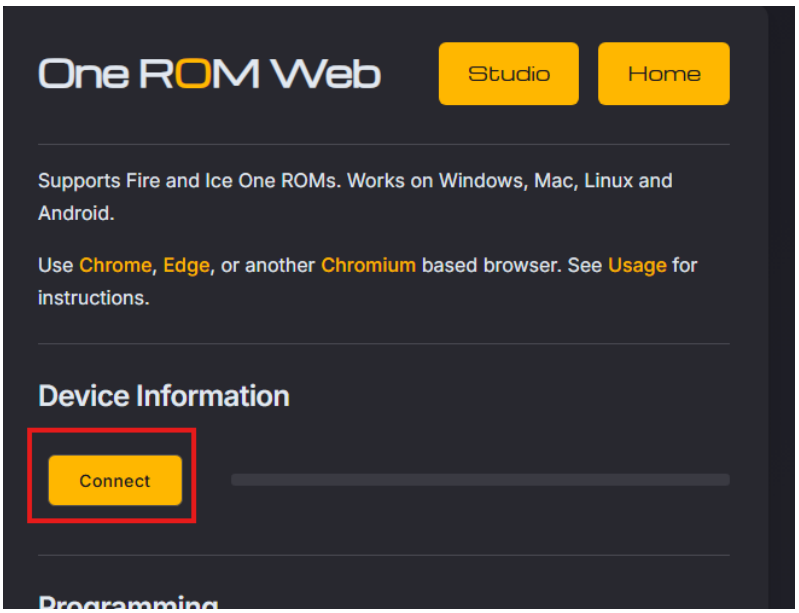
You will need to connect to https://onerom.org/ and you can also find all additional information here https://github.com/piersfinlayson/one-rom/blob/main/docs/GETTING-STARTED.md#multi-rom-sets

In the example below, I am using the Commodore PET 8032 NTSC computer. However, the examples can be used for other computers.

## Using One ROM with one image

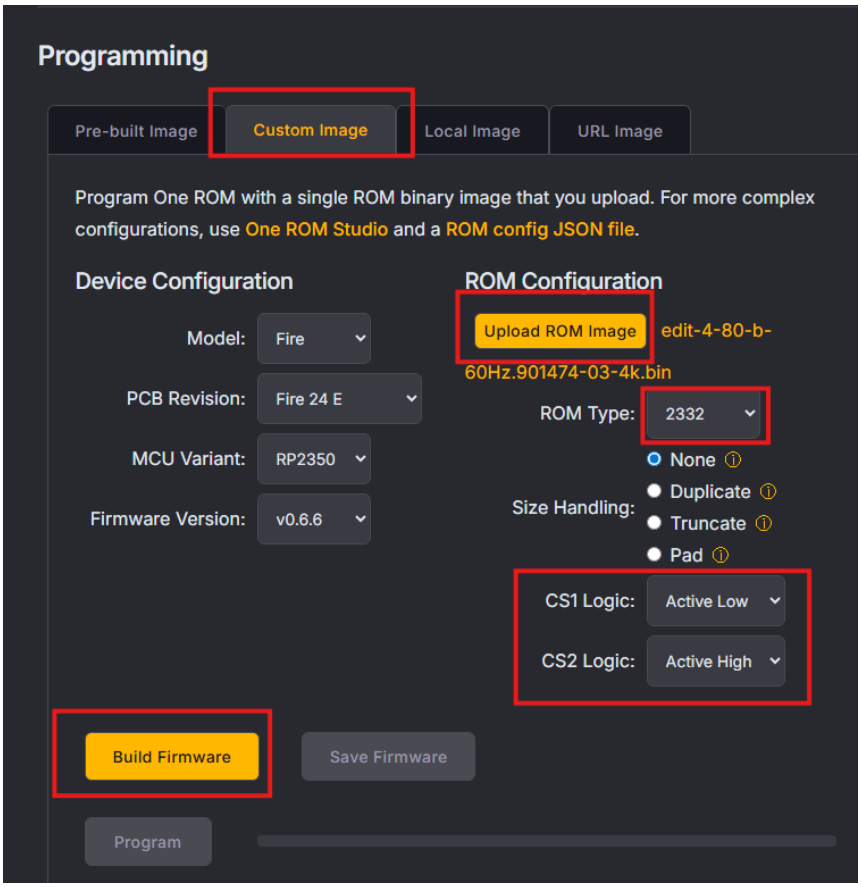
Just load 1 image onto the One ROM then follow these steps:

1. If you just want to load a single ROM, it's much easier to use web: https://onerom.org/web/
2. With the USB cable connected from your computer and the One ROM board, hit "Connect" (it will figure out what sort of One ROM you have and fill in the device config on the left-hand side for you).



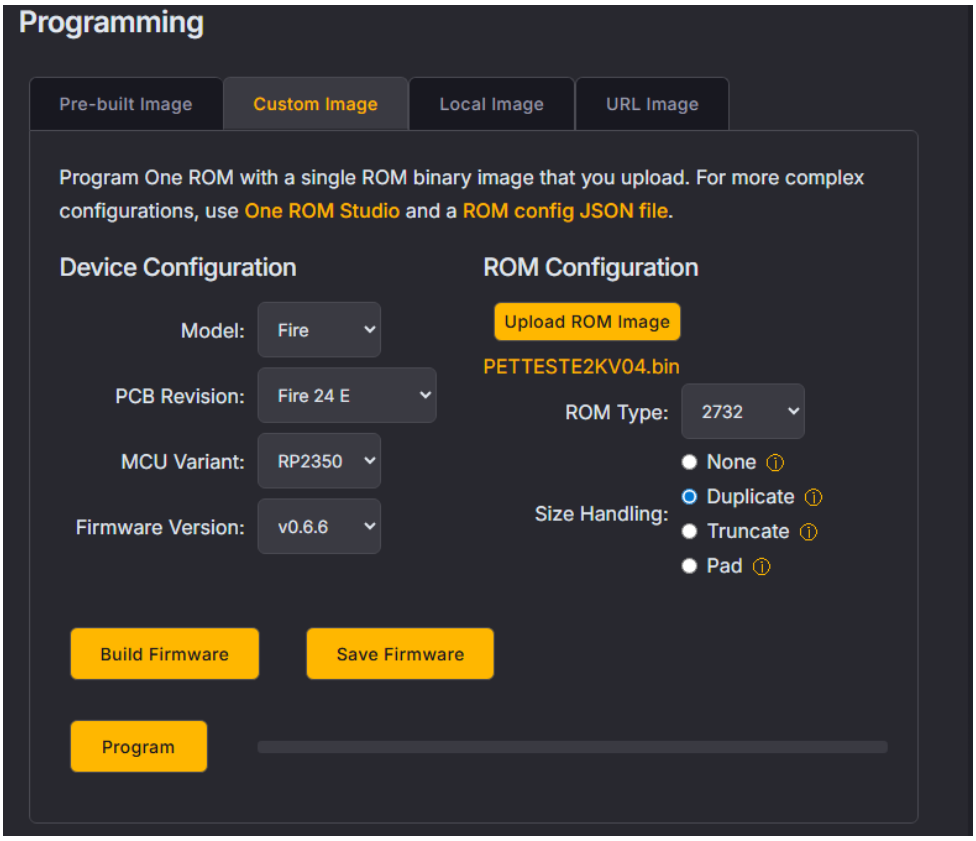

Once connected the information about the One ROM is loaded.

3. Upload ROM image. Select “Custom Image”, then click on “Upload ROM Image”




4. Select “ROM type”
5. If your ROM image and ROM type match size-wise, no need to select anything on size handling,
6. Select “Build Firmware”
7. Once completed, select “Program”
8. Disconnect the USB cable and ready for testing in your computer

This example below is for programming the PETTEST2KV04.bin file with 4KB files size.



# Using One ROM with multiple images

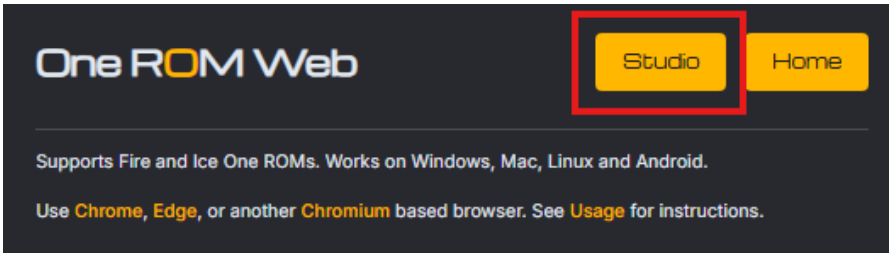
To start, you need:

1. Download the One ROM Studio software and install it. See web page for details.
2. Create a .json file.
3. Connect to One ROM and upload image(s)
4. Selecting the Multiple Images

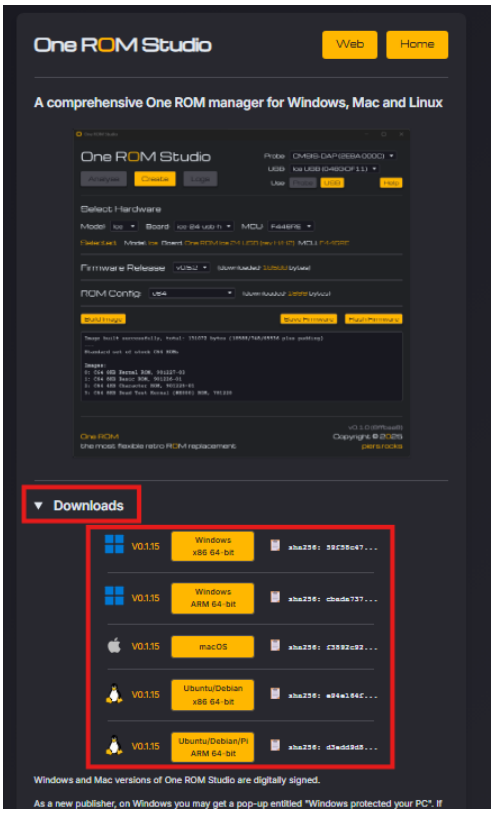
**Download the One ROM Studio software and install it**

https://onerom.org/web/

Click on the Studio button.




Next, scroll down to the “Downloads” section and download the version for your computer. In my example, I am using the Windows version




# Create a .json file

This file contains the configuration for creating multiple images. Here is my .json file that I have for the Commodore 8032 60Hz (NTSC) PET computer.

Before starting here are some notes:

1. If you are using Windows, then note that there are double backslashes:

   "C:****Users****Rudy****Downloads****8032****schema.json"

2. The images (.bin) file are 4K in size. One ROM expects a 4KB image. I used the following:
   - **edit-4-80-b-60Hz.901474-03-4k.bin**
   - **PETTESTE2KV04-4k.bin**

3. The ROM type used is the 2332:

   **"type": "2332"**

4. CS1 an CS2 are set to active low and active high:

   **"cs1": "active_low",**
   **"cs2": "active_high"**

Below is the .json file I used

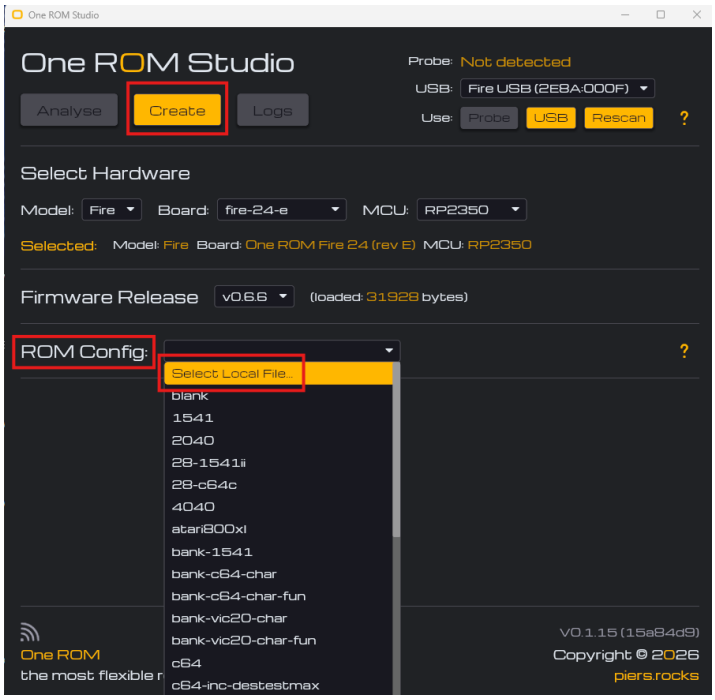

```
{
  "$schema": "C:\\Users\\Rudy\\Downloads\\8032\\schema.json",
  "version": 1,
  "name": "PETXX32",
  "description": "PET PETTester and EDIT ROMs",
  "categories": [
  ],
  "rom_sets": [
    {
      "type": "single",
      "roms": [
        {
          "description": "PET EDIT ROM, 901474-03",
          "file": "C:\\Users\\Rudy\\Downloads\\8032\\edit-4-80-b-60Hz.901474-03-4k.bin",
          "type": "2332",
          "cs1": "active_low",
          "cs2": "active_high"

        }
      ]
    },
    {
      "type": "single",
      "roms": [
        {
          "description": "PETTester ROM, PETTESTE2KV04",
          "file": "C:\\Users\\Rudy\\Downloads\\8032\\PETTESTE2KV04-4k.bin",
          "type": "2332",
          "cs1": "active_low",
          "cs2": "active_high"
        }
      ]
    }
  ]
}
```


Once your file has been updated, save it as a .json file.

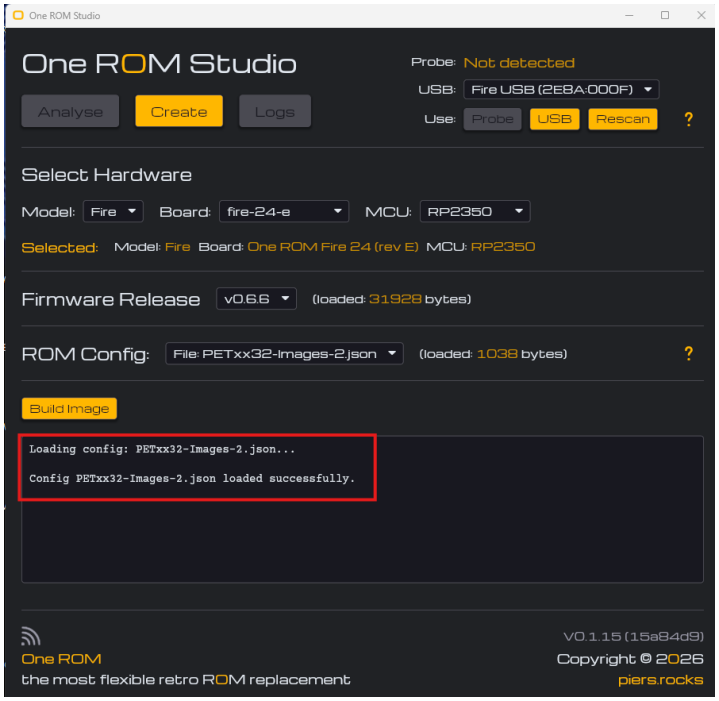
# Connect to One ROM and upload image(s)

Open the One ROM Studio program and click on the “Create” button

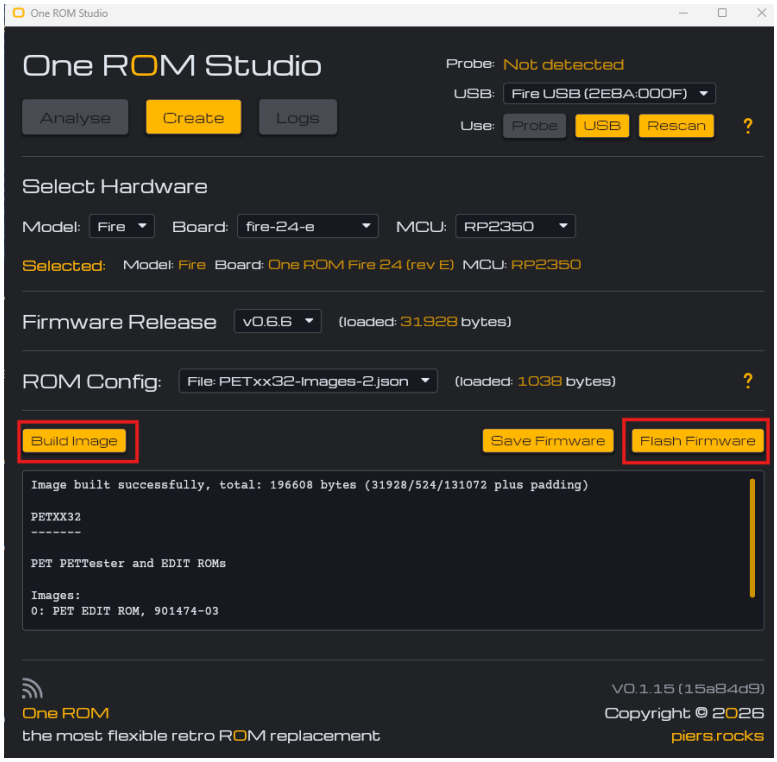
Then in the “Rom Config” section, use the pull down and select “Select Local File”




Next a window will open. Navigate to the location of your newly created .json file and select it.



The .json file loads the new configuration. At this point, once successfully loaded, click on the “Build Image” button

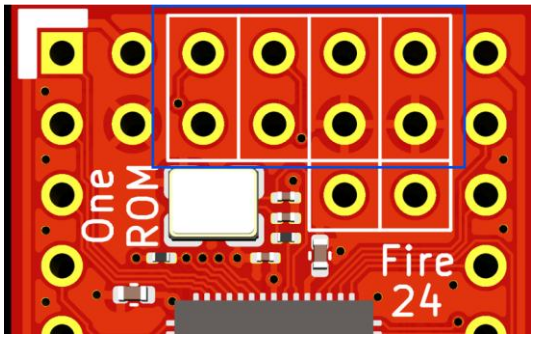



Finally, click on the “Flash Firmware” button. This will send the updated firmware to the One ROM board connected with the USB cable.

# Selecting the Multiple Images

Now that the 2 images have been programmed to your One ROM you will need to set the jumpers so that you can select either image on boot up. Note that the computer must be turned ***OFF*** before switching images.

In this example, the One ROM is the Fire 24 board. See app web page for more details.



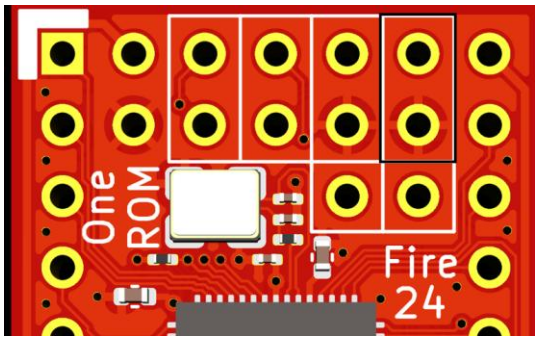

There are 4 rectangular white boxes, I have placed a blue box around the pins.

To select the first image, you just ensure there are no jumpers

First image = 0000 = 0 – default 1st image

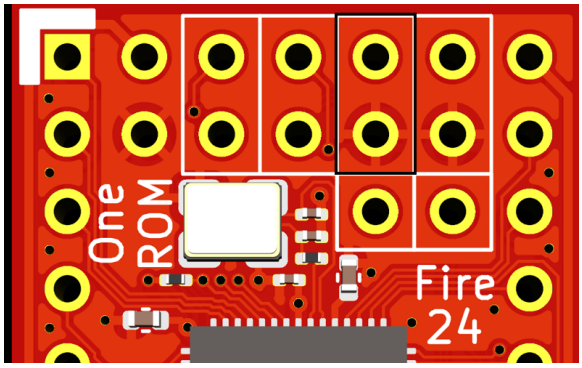
Place a jumper on the send set of pins to access the 2nd image

Second image = 0001 = default to 2nd image



If you have a 3rd image then set the jumper on the 3rd set of pins

Third image = 0010 = defaults to 3rd image

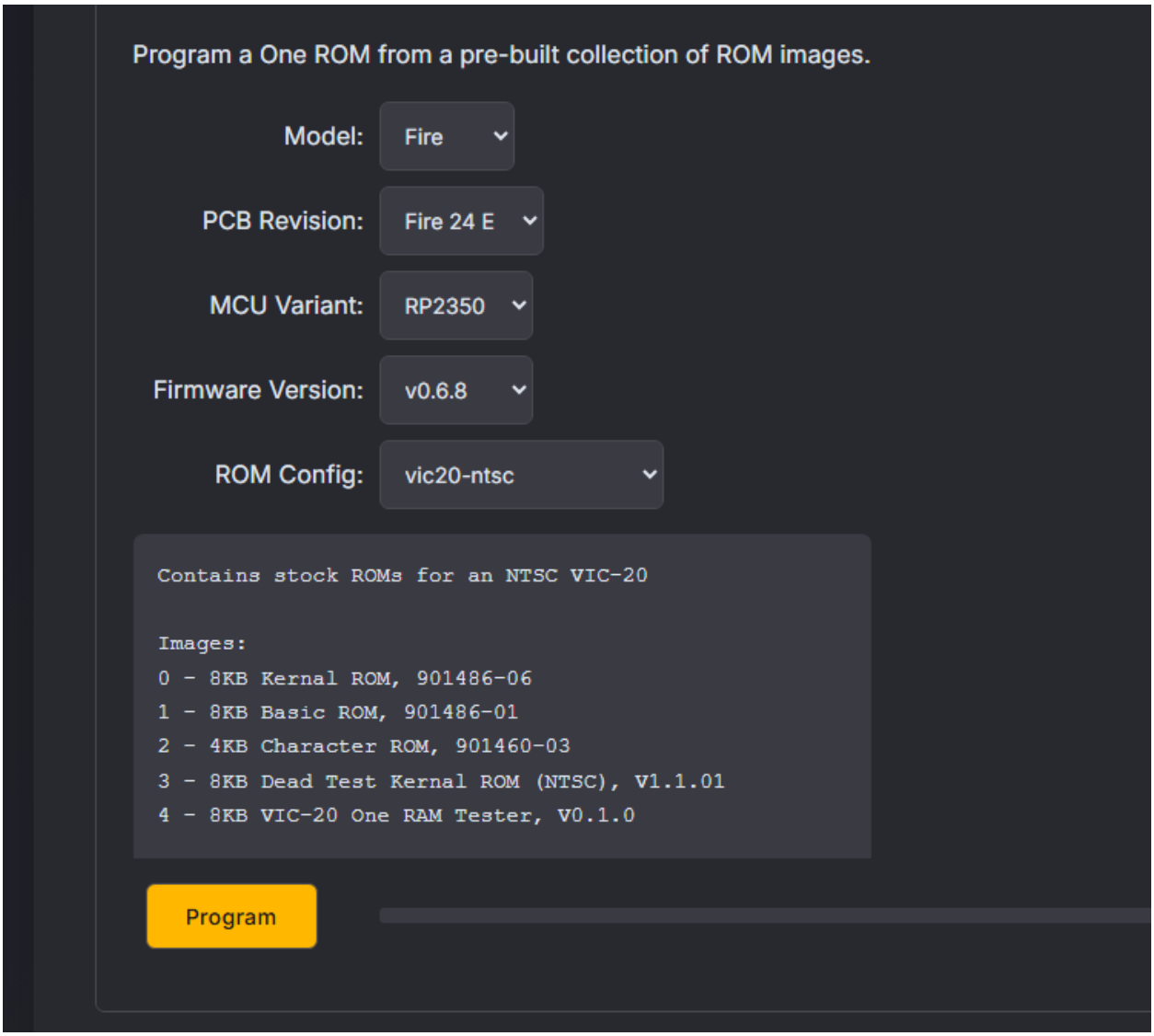



etc.

Finally, boot your computer with either image selected. If it works, turn off your computer and select the other image and turn on your computer. Your 2nd image should be loaded.

# OneROM on the Commodore VIC20 NTSC Computer

Using the OneROM on my NTSC Commodore VIC20, I selected the ROM Config and selected the “vic20-ntsc” and programed the ROM.



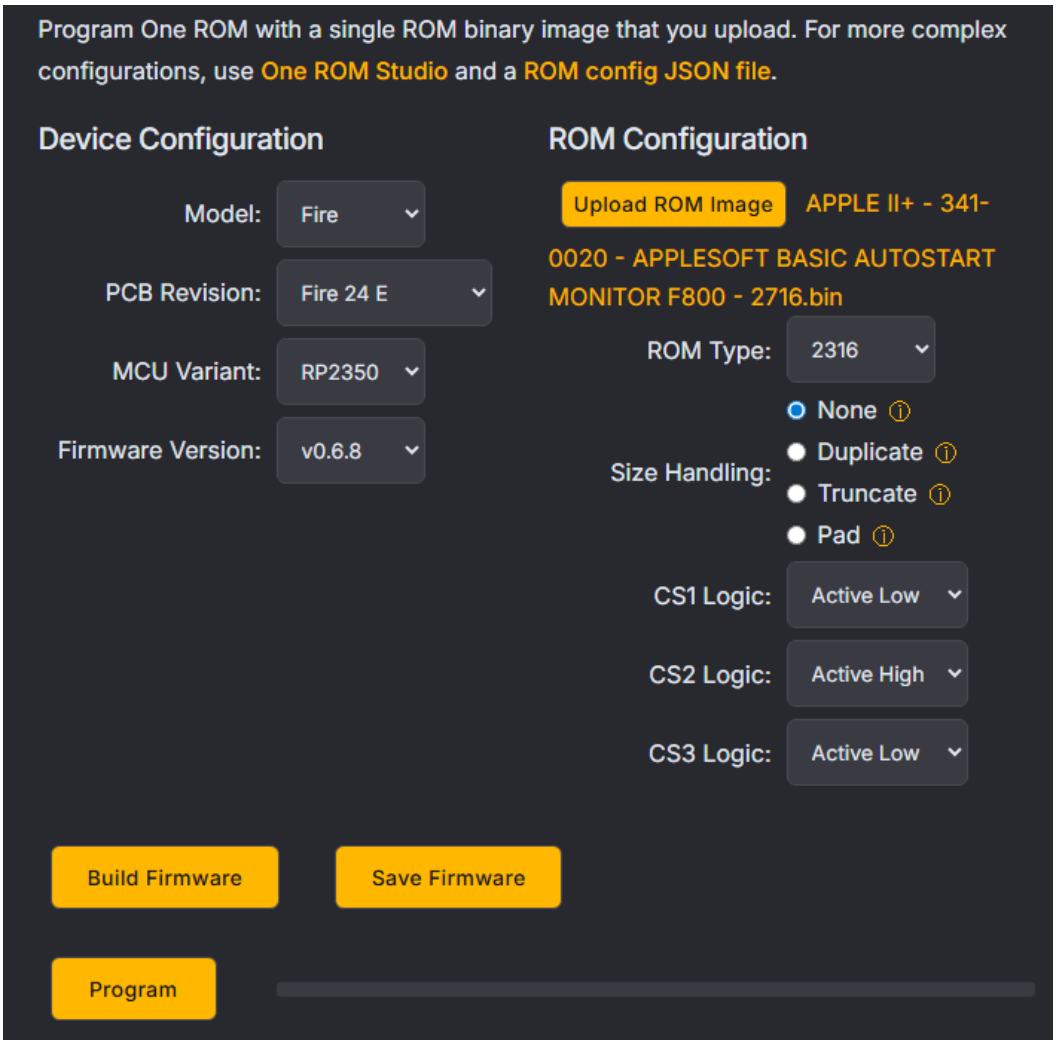

This ROM can now replace the Kernal, Basic or Character ROM. Using the jumpers, see above, you can select which ROM is being replaced. If selecting #3 then the Dead Test will execute on boot up. If selecting #4 then the VIC-20 One RAM Tester will start on boot up.

# OneROM on the Apple II + NTSC Computer with 2 Images

When programing the OneROM for the Apple II+ computer, use the 2316 ROM and select the following settings for CS1, CS2 and CS3

- **CS1** – must be Low (**active-low**)
- **CS2** – must be High (**active-high**)
- **CS3** – must be Low (**active-low**)

In my case I am creating a new F8 ROM for the Apple II+




Program the OneROM and test in the computer

# Using OneROM with 2 Images for the Apple II+ Computer

Here is the .json file needed. Change the path and file names for each image as needed.


```
{
  "$schema": "C:\\Users\\Rudy\\Downloads\\OneROM\\schema.json",
  "version": 1,
  "name": "APPLE II+ F8 ROM - 2316",
  "description": "Apple II+ and DEAD Test ROMs",
  "categories": [
  ],
  "rom_sets": [
    {
      "type": "single",
      "roms": [
        {
          "description": "Apple II+ F8 ROM - 341-0020",
          "file": "C:\\Users\\Rudy\\Downloads\\OneROM\\AppleIIROMs\\APPLE
II+F8ROM.bin",
          "type": "2316",
          "cs1": "active_low",
          "cs2": "active_high",
          "cs3": "active_low"



        }
      ]
    },
```

```
    {
      "type": "single",
      "roms": [
        {
          "description": "Apple II+ DEAD Tester",
          "file": "C:\\Users\\Rudy\\Downloads\\OneROM\\Apple-II-DeadTest-
ROM\\apple2dead.bin",
          "type": "2316",
          "cs1": "active_low",
          "cs2": "active_high",
          "cs3": "active_low"

        }
      ]
    }
  ]
}
```